Sandu Stefan | Cybersecurity & AI Student

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Professional Summary

Second-year Economic Cybernetics student at ASE Bucharest with a security-first approach to software engineering. Competed in two international hackathons in 2026: reverse-engineered a production flight-search API at VibeHack Bucharest (24 h) and built a privacy-preserving AI workspace with local-first data architecture at the AI Agent Olympics during Milan AI Week at Fiera Milano (7-day hybrid build). Combines quantitative training (statistics, econometrics, Python, C++) with hands-on experience in API security analysis, secure credential management, and privacy-by-design systems. Seeking internships in cybersecurity, penetration testing, or AI security where I can apply my builder mindset and analytical rigour.

Education

Academy of Economic Studies (ASE) **Bucharest**
B.Sc. Economic Cybernetics, Statistics & Computer Science *2024 – 2027 (expected)*
Relevant coursework: Statistics, Mathematics, Econometrics, Databases, Programming Fundamentals, Data Security.

Technical Skills

Languages: Python (intermediate), C++, SQL, HTML/CSS, JavaScript/TypeScript (reading level)

Frameworks: FastAPI, Next.js (exposure), Electron (exposure)

Tools: Git/GitHub, Linux, Docker, Twilio API, Google Gemini API, OpenAI API, Firebase

Security: API reverse engineering & traffic analysis, secure credential management, privacy-by-design architecture, local-first data storage patterns

Design: Figma, Adobe After Effects, Premiere Pro, Photoshop, Illustrator

Hackathon Projects

ITFest 2026 – Challenge 4 **Romania**
InvoiceGuard — Pre-Payment Invoice Fraud Detection *24–25 Oct 2026*
Built a desktop app that checks every incoming e-Factura invoice from ANAF's SPV before it is paid or imported into accounting software (SAGA, WinMentor), giving a clear, explained verdict: Verified, Review, or Blocked.

- Designed an explainable **fraud-detection rule engine** that catches changed IBANs (vs. contract and payment history), look-alike "clone" suppliers, duplicate invoices, and unusual amounts, with a 0–100 risk score.
- Integrated the **ANAF public API live** to verify that suppliers exist, are active, match their CUI and are registered for VAT — respecting rate limits (100 CUIs/request, 1 req/s) with a 6-hour cache and offline fallback.
- Built a **hardened XML parser** for the official e-Factura format (UBL 2.1 / CIUS-RO) using defusedxml to block XXE and XML-bomb attacks; validated CUI check digits and IBAN checksums.
- Shipped a cross-platform desktop app (pywebview, Windows/macOS) on a local FastAPI server — all data stays on the accountant's machine, ANAF being the only external call — plus a web demo and a Chrome extension.

Tech: Python, FastAPI, pywebview, JavaScript, HTML/CSS, httpx, defusedxml, ANAF API, Chrome Extension

AI Agent Olympics – Milan AI Week 2026 **Fiera Milano, Italy (in-person)**
PrivAI — Privacy-First AI Workspace *13–20 May 2026*
Built a desktop AI workspace for business automation with a team of 4 at Europe's largest AI conference ($32 000+ prize pool, hosted by lablab.ai). All chat data stays on-device in SQLite; Firebase handles only authentication.

- Developed a **privacy guard** module that applies regex-based redaction to outbound prompts, preventing PII leakage to external LLM providers.
- Built a FastAPI backend serving a Next.js/Electron frontend with SSE streaming, file tools, and terminal access for autonomous agent workflows.
- Implemented **local-first architecture**: conversations, search caches, and redaction logs stored exclusively in on-device SQLite — zero chat data touches the cloud.
- Integrated Google Gemini API with multi-model support (text + vision), orchestrated via an agentic pipeline with tool-calling, web search (SearXNG), and approval workflows.

Tech: Python, TypeScript, FastAPI, Next.js, Electron, Firebase Auth, SQLite, Gemini API

VibeHack Bucharest – Vola.ro AI Challenge **Bucharest**
JetSeeker — WhatsApp AI Travel Assistant *14–15 Mar 2026*
Built a WhatsApp-native AI travel assistant in 24 hours for Romania's largest online travel agency (vola.ro). The bot searches real flights, analyses booking screenshots, and identifies destinations from photos — all inside WhatsApp.

- **Reverse-engineered Vola.ro's production flight-search API**: intercepted network traffic, analysed request/response structures, session handling, and polling mechanisms to build a reliable data integration layer.
- Built an LLM function-calling pipeline (Qwen3-8B) that extracts IATA codes and dates from natural language, calls the flight API, and formats results conversationally.
- Integrated Twilio WhatsApp API with FastAPI, using background task processing to handle Twilio's webhook timeout constraints.
- Designed for **group chat dynamics**: selective response logic (reply only when tagged), multi-turn context management, and handling conflicting travel preferences from multiple users.

Tech: Python, FastAPI, Twilio, OpenAI-compatible API, Vola.ro API (reverse-engineered)

Work & Travel

Frank & Patty's **Virginia Beach, VA, USA**
Multi-Role Restaurant Associate (U.S. Work & Travel) *Jun 2026 – Sep 2026*

- **Server, Food Runner & Busser:** Welcomed guests, took accurate food and beverage orders, answered menu inquiries, delivered dishes promptly, and cleared/reset tables to ensure rapid turnover during peak hours.
- **Line Cook:** Prepared and plated burgers, hot dogs, salads, and desserts in a fast-paced kitchen environment, maintaining quality control and strict sanitation standards.

Ben & Jerry's **Virginia Beach, VA, USA**
Customer Service Associate (U.S. Work & Travel) *Jun 2026 – Sep 2026*

- Provided friendly, fast-paced service to hundreds of customers daily, handcrafting custom sundaes, milkshakes, and smoothies.
- Operated POS cash registers, balanced daily transactions, restocked inventory, and maintained store cleanliness.

Experience

AdSymphony **Bucharest**
Social Media Marketing Assistant *Oct 2022 – Nov 2022*

- Designed UX/UI prototypes for client websites using Figma, improving layout clarity and user flows.
- Created motion graphics and animated social media content in Adobe After Effects, boosting campaign engagement.
- Managed design file libraries and iterated on deliverables based on real-time client feedback.

Emblema **Bucharest**
Retail Sales Assistant *Jun 2022 – Aug 2022*

- Assisted customers with product selection and maintained a well-organised store floor, contributing to a positive shopping experience.

Activities & Competitions

SCOALA-AMBASADOR – EU Programme **Bucharest**
Volunteer Ambassador *Mar – May 2022*
Organised school activities supporting the EU School Ambassador initiative, promoting European values and civic engagement.

Erasmus+ IDEAS Project **Bucharest**
Digital Skills Trainer *Oct 2022*
Welcomed and mentored visiting teachers and students from EU partner countries, delivering hands-on digital skills workshops.

ASUS Learn & Compete 2023 **Bucharest**
Competitor *Apr 2023*
Competed in a national IT knowledge competition, demonstrating proficiency in computer science topics.

Volunteer Work

Neghiţă Children's Centre **Bucharest**
Volunteer *Sep 2021 – May 2024*
Supported activities for underprivileged children over three years, demonstrating sustained commitment, empathy, and social responsibility.

Languages

Romanian (Native) English (Professional – B2/C1) French (B1)